

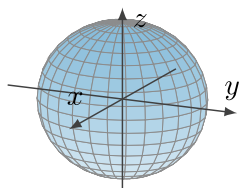
# Assignment 6

Name: \_\_\_\_\_

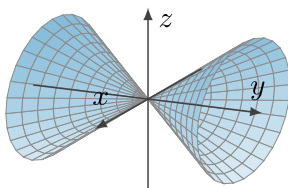
Student ID: \_\_\_\_\_

8 problems. Show all of your work.

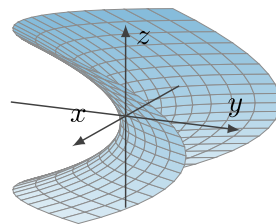
## Problem 1. Matching Quadric Surfaces



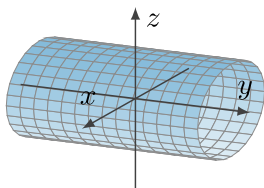
**A**



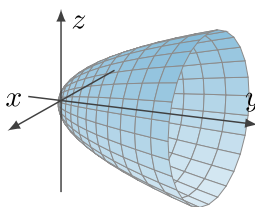
**B**



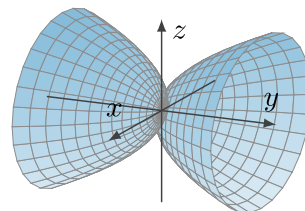
**C**



**D**



**E**



**F**

Match the equations of the surfaces with the graphs A–F shown above by entering a letter from A to F in each blank.

$y = 2x^2 + z^2$  matches graph \_\_\_\_\_.

$y^2 = x^2 + 2z^2$  matches graph \_\_\_\_\_.

$x^2 + 2z^2 = 1$  matches graph \_\_\_\_\_.

$y = x^2 - z^2$  matches graph \_\_\_\_\_.

**Problem 2.**

Identify and sketch the surface corresponding to the following equation:

$$z^2 - y^2 - x^2 = 0.$$

**Problem 3.**

Identify and sketch the surface  $x^2 - y^2 + z^2 - 4 = 0$ .

**Problem 4.**

Identify and sketch the surface corresponding to the equation  $2x^2 - y + z^2 = 0$ . Give reasons for your choice.

**Problem 5.**

Sketch the surface  $4x^2 - y^2 = -4 - z^2$  in three dimensions. Label any points where the surface crosses the coordinate axes.

**Problem 6.**

Find the equation of the hyperbola with asymptotes  $y = 3x$  and  $y = -3x$  passing through the point  $(0, 2)$ .

**Problem 7.**

Find the equation of the parabola with vertex at the origin and axis of symmetry the  $y$ -axis, passing through the point  $(4, 2)$ . Then find its focus and its directrix.

**Problem 8.**

Two trees grow 100 feet apart. A 120 foot rope is attached to the first tree at a height of 100 feet and to the second tree at a height of 50 feet. A child folds an empty plastic water bottle over the rope and rappels from the high end of the rope to the low end. At each moment, the rope is taut. The point where the water bottle is folded over the rope traces a path. Ignoring the numbers, state what kind of curve the path makes: line, ellipse, parabola or hyperbola.